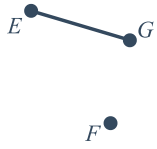


Line segments and distance



1



2 $\overline{AC}$

3 $\overline{XW}$

4 $\overline{AE}$

5 $\overline{DF}$

6

a 5 units

b 5 units

7

a 4 units

b 1 unit

c $c = \sqrt{17}$

8

a 2 units

b 1 unit

9 $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

10 Distance = $\sqrt{(-5 - 4)^2 + (10 - (-5))^2}$

11 10.00

12

a 15 units

b $\sqrt{73}$ units

13 $\sqrt{34}$ units



14

- a $\sqrt{82}$ units
- b $\sqrt{26}$ units
- c N

15

- a $\sqrt{50}$ units
- c M

b 5 units

16

- a $\sqrt{50}$ units
- b $\sqrt{2}$ units
- c 5 : 1

17

a $CE = 28$

b $GH = 15$

18

a $QS = 63$

b $PQ = 24$

19

a $x = 3$

b 25

20 $JT = 40$

21

- a
 - i $x = 3$
 - ii 6
- b
 - i $x = 3$
 - ii 3



Additional questions

22

a

- i $AB = 3$ units
- ii $BC = 4$ units
- iii $AC = 5$ units

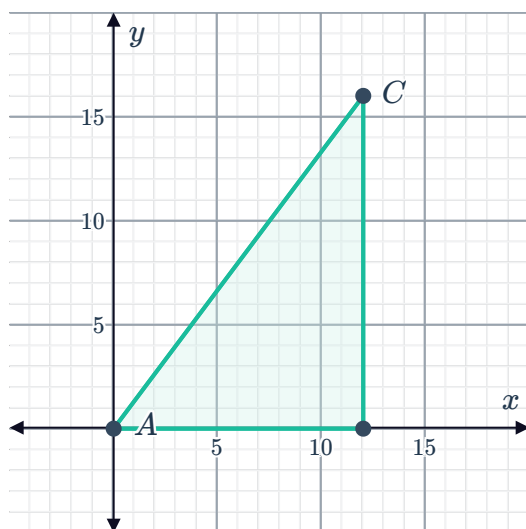
b

- i $AB = 4$ units
- ii $BC = 7$ units
- iii $AC = 8.06$ units

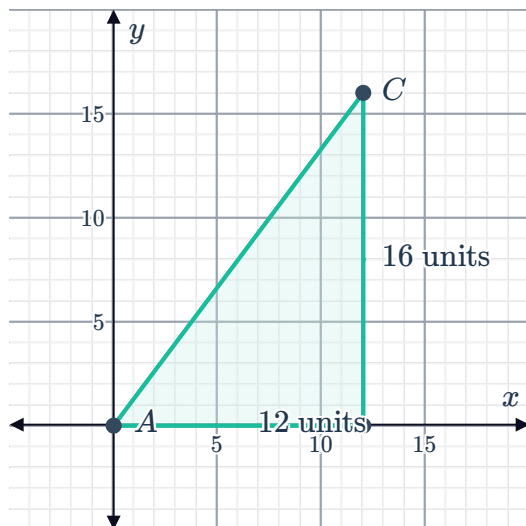
23

a

i



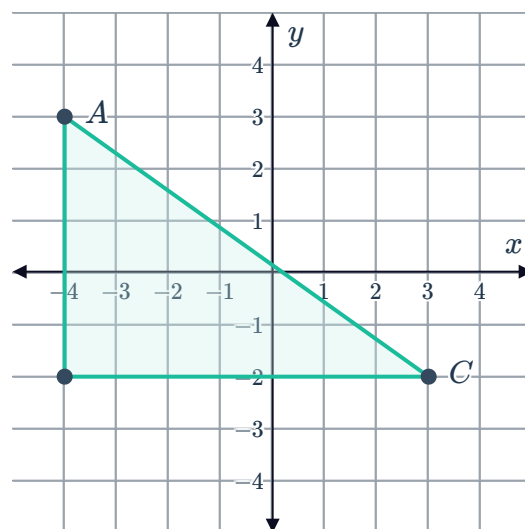
ii



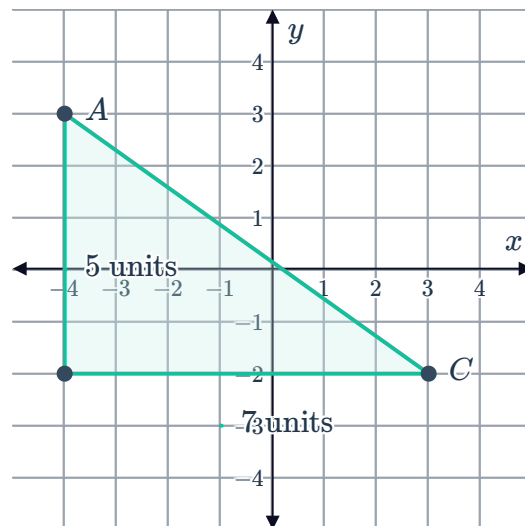
- iii $AC = 20$ units

b

i



ii



- iii $AC = 8.6$ units

24



25

- a 10 units
- b 15 units
- c 8.54 units
- d 5.83 units

26

- a $PM = 9.90$ units
- b $PN = 11.18$ units
- c N

27 17 units

28

- a $AC = 14.14$ units
- b $BD = 8.49$ units

29

- a Culmer Station is the closest to their meeting spot.
- b When actually walking, we need to consider traffic lights, side walks and what is accessible. Depending on which direction they are catching the train may also impact their choice.
- c If they need to stop at the Winn-Dixie, then the Civic Center Station would have the shortest trip distance.

30

- a There are two possible coordinates.
- b $M(-2, 9)$ and $M(-2, 3)$
- c If we were not told the line segment was vertical, there would be an infinite number of answers.

31 The distance formula comes from the Pythagorean theorem. $(x_2 - x_1)$ is the calculation for the length of the line segment which is the horizontal component of the right triangle we can create. $(y_2 - y_1)$ is the calculation for the length of the line segment which is the vertical component of the right triangle we can create.

32 Both Dayanna and David are correct. $(x_2 - x_1)$ and $(x_1 - x_2)$ will have the same magnitude, but different signs. However, when we square them, they will both become positive, so they will give the same answer.

33



- a Fyfe made an error on Step 2. They incorrectly substituted $(x_2 - y_2)$ and $(y_1 - x_1)$, but this calculation does not connect to the Pythagorean theorem to make a right triangle and so does not find the distance.

b

$$\begin{array}{ll}
 \textcircled{1} & AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad \text{Distance formula} \\
 \textcircled{2} & = \sqrt{(3 - -4)^2 + (-5 - 7)^2} \quad \text{Substitution Property of Equality} \\
 \textcircled{3} & = \sqrt{(7)^2 + (-12)^2} \quad \text{Simplify} \\
 \textcircled{4} & = \sqrt{49 + 144} \quad \text{Simplify} \\
 \textcircled{5} & = \sqrt{193} \quad \text{Simplify}
 \end{array}$$

34

- | | | | |
|---|----------|---|----------|
| a | $AC = 1$ | b | $AE = 4$ |
| c | $EF = 1$ | d | $FB = 2$ |
| e | $DB = 4$ | f | $AB = 7$ |

35

- | | | | |
|---|--|---|--|
| a | By the segment addition postulate we know that $CD + DE = CE$ and so $CE = 28$ | b | By the segment addition postulate we know that $FG + GH = FH$ and so $GH = 15$ |
|---|--|---|--|

36

- | | | | |
|---|-----------------------|---|------------------------|
| a | $PR = 5.1 \text{ mm}$ | b | $AB = 51.4 \text{ cm}$ |
|---|-----------------------|---|------------------------|

37

- a $\overline{DE}$ and $\overline{EF}$
- b $\overline{CF}$ and $\overline{DB}$
- c In part (a) you would find two new congruent segments, $\overline{CG}$ and $\overline{DG}$, and in part (b) you would not find any additional congruent segments.

38

a $KL = 25$

b $JT = 40$



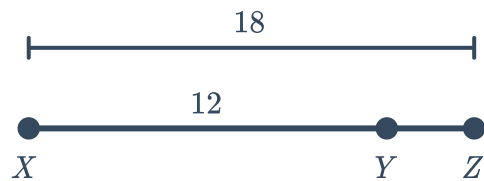
39

a $BD = 12$

b $CB = 3$

40 $ML = 51$

41 YZ could be 30 units in length or 6 units in length:



42 By the segment addition postulate, if points A , B , and C were collinear, then $AB + BC = AC$.

Since $AB + BC = 15$ and $AC = 12$, we know that these points cannot be collinear.